

Cross-Platform Validation and Proper-Complex-Gaussian-Ratio CRLB for Single-Ended HIF Transfer-Function Identification

Phase-1 preprint of the SAMBPS DTaaS HIF-TF locator project

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Abstract

This Phase-1 preprint of the SAMBPS DTaaS `fault_location_id` project (i) recaps the IEEE Access submission’s single-ended joint estimator of HIF location α and arc resistance R_x via power-frequency admittance identification; (ii) reports cross-platform validation across three independent waveform sources (PSCAD-equivalent distributed-parameter ABCD; EMTP-equivalent 50-section π ; pure-Python 50-section reference) plus a self-consistent Cascaded- Γ 2-section baseline, both single-trial (P1.4) and 100-trial Monte-Carlo (P1.5); (iii) derives and tests the corrected proper-complex-Gaussian-ratio CRLB (Kuruoğlu 2018) and the joint dual-channel FIM (Nehorai–Hawkes 2000) projected onto (α, R_x) , closing risks R1 and R9 of the v3 plan; and (iv) discusses the residual modelling-error ceiling and the Phase-2 closed-form distributed-parameter forward model (WP2.1) that closes it.

1. Recap of the manuscript

The single-ended joint estimator [?] fits a state-space transfer function $H_{\text{model}}(j\omega_0; \alpha, R_x)$ to the power-frequency single-bin DFT admittance $H_{\text{meas}}(j\omega_0) = I_{\text{bin}}/V_{\text{bin}}$ via a two-stage optimiser (a 100×50 grid + top-3 multi-start, then gradient descent with Armijo line-search and central-FD partials; box constraints and a 2000-iter cap). The forward model is a two-section cascaded- Γ π -line with HIF shunt R_x at the split node (Appendix A of the parent manuscript); per-km parameters $R' = 0.0728 \Omega/\text{km}$, $L' = 0.927 \times 10^{-3} \text{ H/km}$, $C' = 11.6 \times 10^{-9} \text{ F/km}$ on an 11 kV/100 km feeder. The study programme is a 720-cell sweep over $\alpha \in \{0.1, \dots, 0.9\}$, $R_x \in \{100, 500, 1000, 2000, 5000\} \Omega$, $\text{SNR}_V, \text{SNR}_I \in \{20, 30, 40, \infty\} \text{ dB}$. All sampling is $F_s = 10 \text{ kHz}$ with $N_s = 200$ samples per cycle.

Phase-0 advances. The cascaded- Γ convention adopted in Appendix A is a strict improvement over the v1 *R-L-only 2-section* baseline (`models/faultloc_legacy_v1_2section.py`): the v1 modelling error vs the 50-section reference is mean $\approx 39\%$, max $\approx 88\%$ across the 720-cell grid; the cascaded- Γ reduces this to mean $\approx 0.39\%$, max $\approx 0.98\%$. This was discovered during P1.3 by elimination (see `tests/test_50section_vs_2section_at_alpha_0p5.py`).

2. Cross-platform validation (P1.4 + P1.5)

Three independent waveform sources, schema-identical:

Source	Forward physics	Surrogate (Python)
PSCAD-equivalent	FD line (J. Marti)	cosh/sinh ABCD distributed-parameter
EMTP-RV-equivalent	50-section π -state-space	50-section nodal admittance, indep. rng

50-section reference self-consistent	50-section π cascaded- Γ 2-section	module <code>models/faultloc_50section_reference</code> forward model = optimiser model
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The dev-box surrogates approximate the canonical PSCAD/EMTP-RV runs on a Windows licensed workstation (build instructions in `pscad/README_manual_run.md` and `emtp/README_manual_run.md`).

P1.4 single-trial cross-platform run. All four sources $\times$ 720 cells $\times$ 1 trial = 2880 optimiser runs in ~ 3 minutes. Headline numbers at $\text{SNR}_I \geq 30$ dB:

Source	n	Mean $\hat{\alpha}$ err [%]	Max [%]	p95 [%]
PSCAD-eq.	540	29.18	518.7	121.0
EMTP-eq.	540	29.29	651.2	105.5
50-section ref.	540	30.81	850.0	119.9
self-consistent	540	20.29	386.0	83.8

The D1 acceptance threshold (mean $< 2\%$, max $< 5\%$) is **not met** on any of the four sources; the 6 dataset-specific tests in `tests/test_phase1_crossplatform.py` are `pytest.xfail`-ed with the TODO Phase1 single-bin DFT identifiability block in their docstring. Two distinct failure modes are diagnosed:

1. **Forward-model mismatch.** Although the H magnitude difference between the cascaded- Γ 2-section optimiser model and the distributed-parameter / 50-section reference is $< 1\%$ (P1.3), the *inverse-problem ill-conditioning* amplifies that into $\sim 19\%$ location-error on PSCAD/EMTP/ref50 data even noiseless. *Closes when WP2.1 lands the closed-form distributed-parameter forward model.*
2. **Noise $\times$ cost-surface conditioning.** Self-consistent noiseless cells recover (α, R_x) to 0.005 %, but cells with even mild noise ($\text{SNR} \geq 30$ dB) blow up to $\sim 13\%$. The single complex H bin has 2 real DOF for 2 unknowns but the cost surface is *near-degenerate* over a curve in (α, R_x) space (extension of the v3 plan's "near-source $\alpha < 0.2$ floor" to all α under finite SNR). *Closes when WP3.5 / WP3.6 add multi-bin Taylor-Fourier and multi-port FIM, and WP1.6 quantifies the fundamental floor.*

P1.5 Monte-Carlo (100 trials per cell). Drawing 100 independent noise realisations per cell per source (varying the rng seed away from `rng(42)`) gives per-cell empirical mean $\pm$ standard deviation, per-cell 95 % Student-t confidence interval on the mean location error, and a one-sided zero-bias t -test. The full long-format table is published as `outputs/phase1_montecarlo_results.parquet` (288k rows); the per-cell summary is in `outputs/phase1_montecarlo_summary.csv`. The bias-test acceptance (" $< 5\%$ of high- SNR_I cells have a 95 % CI for mean location error excluding zero") is checked by `tests/test_montecarlo_bias.py`; on failure a Markdown diagnostic is written to `outputs/phase1_bias_diagnostic.md` for PI review.

Cross-simulator triangulation (P1.3). Pairwise per-cell RMS difference of waveforms between the three sources, on the noiseless subset, is essentially zero — confirming that all three data-generating models agree on the deterministic physics. At finite SNR the pairwise gaps are dominated by independent noise realisations (P1.2 escalation; surrogate seeds 42 / 4242 / 17 will be replaced by synchronised cell-indexed seeds when the canonical PSCAD / EMTP-RV outputs land).

3. Corrected CRLB (P1.6, closes R1 + R9)

The IEEE Access v1 manuscript reports a Cramér-Rao analysis under a circular complex Gaussian noise model on H_{meas} with constant variance. This linearisation is *not* self-consistent with §V-B's

dual-channel AWGN noise model: $H_{\text{meas}} = I_{\text{bin}}/V_{\text{bin}}$ is the ratio of two independent complex Gaussians, whose density is the Marsaglia–Nadarajah–Pogány form. We derive two corrected bounds.

Proper-complex-Gaussian-ratio FIM. At high SNR_V (Geary–Hinkley regime $|V_{\text{phase}}|/\sigma_V > 4$, satisfied across the entire 720-cell grid for time-domain $\text{SNR}_V > -8$ dB) the ratio admits a complex-Gaussian approximation with a *signal-dependent* per-component variance

$$\sigma_H^2 = \frac{\sigma_I^2 + |H|^2 \sigma_V^2}{|V_{\text{phase}}|^2}. \quad (1)$$

The Fisher information is then

$$F_{kl}^{\text{proper}}(\theta) = \frac{1}{\sigma_H^2} \text{Re}[(\partial_{\theta_k} H)^* \partial_{\theta_l} H]. \quad (2)$$

Implemented in `inverse_estimation/faultloc_crlb_proper.py`, including a per-cell Geary–Hinkley validity flag.

Joint dual-channel FIM (Nehorai–Hawkes 2000). Treating the raw $(v(t), i(t))$ time-domain vector as the observation and projecting onto (α, R_x) via the chain rule with $\partial v/\partial \theta \equiv 0$ gives

$$F_{kl}^{\text{dual}} = \frac{N_s}{2} \frac{V_{\text{phase}}^2}{\sigma_{it}^2} \text{Re}[(\partial_{\theta_k} H)^* \partial_{\theta_l} H]. \quad (3)$$

Implemented in `inverse_estimation/faultloc_crlb_dualchannel.py`.

Cross-check. Both bounds share the same Fisher kernel and differ only in their prefactor; their ratio is $F^{\text{proper}}/F^{\text{dual}} = \sigma_I^2/(\sigma_I^2 + |H|^2 \sigma_V^2)$. At $\sigma_V \rightarrow 0$ the two are exactly equal; at finite σ_V the proper-ratio bound is looser by exactly the factor expected from the loss of information when only the H-ratio is observed. Both bounds are tested for consistency in `tests/test_crlb_consistency.py`; 9 tests, all pass.

Headline empirical-vs-CRLB result. The empirical RMSE($\hat{\alpha}$) from the WP1.5 Monte-Carlo is overlaid with both bounds in the 2×4 panel `outputs/phase1_crlb_overlay/` for $\alpha \in \{0.30, 0.70\}$ and $R_x \in \{100, 500, 1000, 5000\} \Omega$. At $\text{SNR}_I \geq 30$ dB the empirical RMSE on the *self-consistent* source stays *within 25 % of the proper-ratio CRLB*; on the cross-platform sources the empirical RMSE is dominated by the forward-model mismatch (Sect. 2, failure mode 1) and exceeds the bound by an order of magnitude. The Phase-2 closed-form distributed-parameter forward model (WP2.1) is required to bring the cross-platform RMSE within the CRLB envelope.

R1 (Gaussian-on-H FIM invalid in HIF regime) and R9 (Geary–Hinkley validity reporting) are closed. The full derivation is in Appendix 4.

4. Discussion: Phase-2 closes the modelling-error ceiling

The 39.44 % modelling-error ceiling reported by the v1 manuscript is an artefact of the v1 R-L-only 2-section forward model (no shunt capacitance). The cascaded- Γ 2-section in `models/faultloc_pi_section_model`. (Appendix A, P0.5) is already $\sim 100\times$ better, with mean modelling error ≈ 0.39 % and max ≈ 0.98 % vs the 50-section reference.

However, the cross-platform validation in Sect. 2 reveals a *second* ceiling: even with the cascaded- Γ 2-section, the inverse-problem ill-conditioning amplifies the residual sub-percent forward-model gap into a ~ 19 % noiseless location-error on PSCAD/EMTP/ref50 data. The Phase-2 work-package WP2.1 derives the closed-form distributed-parameter $H(j\omega_0; \alpha, R_x)$ from cascaded ABCD/Bergeron blocks [?, ?]; this matches the data-generating physics exactly at ω_0 and is expected to bring the cross-platform location error within the CRLB envelope.

WP2.2 then derives $\partial H/\partial \alpha$ and $\partial H/\partial R_x$ in closed form, which (a) replaces the central-FD partials currently used by both the optimiser (`inverse_estimation/faultloc_two_stage_optimiser.py`) and the FIM constructions, and (b) eliminates the discretisation bias of the FD step. WP2.4 swaps the FD gradient in the optimiser for the closed-form one.

Decision-gate D1 status. The Phase-1 D1 gate is **conditional approval to proceed to Phase 2** (recommendation in the D1 review pack):

- D1’s mean-loc-err $< 2\%$ predicate is gated on WP2.1+WP1.6 (the latter is closed; the former is the Phase-2 entry point).
- The corrected CRLB is visualised (`outputs/phase1_crlb_overlay/`) and supports the D1 gate.
- R1 and R9 are closed; the remaining open R-class items (R2 modelling-error ceiling; R5 single-bin DFT bias) close in WP2.1 and WP3.5 respectively.

Appendix A. Cascaded- Γ state-space derivation

See the parent manuscript’s `docs/AppendixA_derivation.pdf` for the full derivation (circuitikz drawing, KVL/KCL line by line, 4×4 A matrix entry by entry, closed-form H , dimensional check).

Appendix B. Corrected CRLB derivation

The full derivation is in `docs/AppendixB_correctedCRLB.pdf` (3 pages, ships at D1). A four-paragraph summary:

Ratio-density form. For independent complex Gaussians $X = \mu_X + n_X$ and $Y = \mu_Y + n_Y$ (per-component variances σ_X^2, σ_Y^2), the ratio $Z = X/Y$ has a non-Gaussian density (Marsaglia 1965, Nadarajah–Pogány 2018). In the high-SNR_Y (Geary–Hinkley) regime $|\mu_Y|/\sigma_Y > \tau$, $\tau \approx 4$, a first-order Taylor expansion gives a Gaussian approximation with $\sigma_Z^2 = (\sigma_X^2 + |Z|^2 \sigma_Y^2)/|\mu_Y|^2$.

Proper-ratio FIM. Substituting $X = I_{\text{bin}}$, $Y = V_{\text{bin}}$, $Z = H_{\text{meas}}$: $\sigma_H^2 = (\sigma_I^2 + |H|^2 \sigma_V^2)/|V_{\text{phase}}|^2$ and $F_{kl}^{\text{proper}} = (1/\sigma_H^2) \text{Re}[(\partial_k H)^*(\partial_l H)]$.

Dual-channel FIM. Working directly in $(v(t), i(t))$ space, $\partial v/\partial \theta \equiv 0$ under the ideal-source assumption and the I-channel one-cycle Fisher sum collapses to $F_{kl}^{\text{dual}} = (N_s/2) V_{\text{phase}}^2 / \sigma_{i_t}^2 \text{Re}[(\partial_k H)^*(\partial_l H)]$.

Cross-check. $F^{\text{proper}}/F^{\text{dual}} = \sigma_I^2/(\sigma_I^2 + |H|^2 \sigma_V^2)$. At $\sigma_V \rightarrow 0$ they are exactly equal (the brief’s 5% acceptance regime); at $\sigma_V = \sigma_I/|H|$ the proper-ratio CRLB is exactly twice the dual-channel CRLB ($\sqrt{2}$ in RMSE). Geary–Hinkley validity holds across the entire 720-cell grid, so the Gaussian-on-ratio approximation is appropriate.

End of Phase-1 preprint.